

Q1. Define ASP.NET. Explain its features.

- ASP.NET is a web development framework developed by Microsoft.
- Used to build dynamic websites, web applications, and services.
- Runs on the .NET platform and supports multiple languages (C#, VB.NET).
- Features:
 - - Server-side programming (code runs on server).
 - - Works under CLR providing type safety and memory management.
 - - Supports Web Forms (drag-and-drop) and MVC pattern.
 - - Cross-platform support via ASP.NET Core.
 - - Integrated with ADO.NET for database operations.
 - - Provides strong security and scalability.

Q2. Explain the ASP.NET Page Life Cycle with events.

- The Page Life Cycle describes how a page is processed from request to response.
- Phases:
 - - Page Request: Check if page needs compilation.
 - - Start: Properties like Request, Response set.
 - - Initialization: Controls are initialized, IDs set.
 - - Load: Postback data loaded, control values restored.
 - - Validation: Validation controls check inputs.
 - - Postback Event Handling: Events like button click executed.
 - - Rendering: Page output generated and sent to browser.
 - - Unload: Resources cleaned, page disposed.
- Important Events: PreInit, Init, PreLoad, Load, Control Events, PreRender, Render, Unload.

Q3. Differentiate between Web Forms and MVC in ASP.NET.

- Web Forms:
 - - Event-driven programming model.
 - - Uses ViewState to maintain state.
 - - Less control over generated HTML.
 - - Suitable for rapid application development.
- MVC:

- - Separates application into Model, View, Controller.
- - No ViewState, provides full HTML control.
- - More testable and maintainable.
- - Suitable for large-scale, modern applications.

Q4. Explain the .NET Framework architecture.

- Components of .NET Framework:
- - CLR (Common Language Runtime): Executes managed code.
- - BCL (Base Class Library): Provides reusable classes (IO, collections, XML).
- - CLS (Common Language Specification): Ensures cross-language interoperability.
- - ADO.NET: Provides data access mechanism.
- - Windows Forms, Web Forms, Web Services for UI and communication.

Q5. What is CLR? Explain its functions.

- CLR (Common Language Runtime) is the execution environment of .NET.
- Functions:
- - Memory Management (Garbage Collection).
- - Type Safety and Code Verification.
- - Exception Handling.
- - Thread Management for multitasking.
- - Security (code access security).
- - Interoperability with COM components.
- - Just-In-Time (JIT) compilation of MSIL to native code.

Q6. Explain Server Controls in ASP.NET with examples.

- Server Controls are objects that run on the server and render as HTML.
- Examples:
- - Label: Displays text (read-only).
- - TextBox: Accepts user input (SingleLine, MultiLine, Password).
- - Button: Executes events (Click event).
- - CheckBox/RadioButton: Selection controls.
- - GridView: Displays tabular data with paging and sorting.
- All server controls must include `runat='server'`.

Q7. What isPostBack and AutoPostBack in ASP.NET?

- PostBack:
 - - A request sent from client to server when page is submitted.
 - - The page is processed again and reloaded.
- AutoPostBack:
 - - A property of controls like DropDownList, CheckBox.
 - - Automatically posts page to server when value changes.
 - - Helps update UI without waiting for button click.

Q8. Explain Data Binding in ASP.NET.

- Data Binding connects controls with data sources.
- Types:
 - - Single-value Binding: Label, TextBox.
 - - Multi-value Binding: GridView, ListBox, Repeater.
- Methods:
 - - Declarative: Using SqlDataSource, ObjectDataSource.
 - - Programmatic: Binding through code-behind using DataBind().

Q9. Explain ADO.NET and its architecture.

- ADO.NET is used for data access in .NET applications.
- Two Architectures:
 - 1. Connected Architecture:
 - - Uses SqlConnection, SqlCommand, SqlDataReader.
 - - Requires active connection, faster and lightweight.
 - 2. Disconnected Architecture:
 - - Uses DataSet, DataTable, DataAdapter.
 - - Works offline, supports multiple tables and relations.

Q10. Differentiate between DataReader and DataSet.

- DataReader:
 - - Connected, forward-only, read-only.
 - - Faster and memory efficient.

- - Does not support XML.
- DataSet:
- - Disconnected, can hold multiple tables.
- - Editable and supports relations.
- - Can be stored in XML format.

Q11. Explain State Management techniques in ASP.NET.

- State Management preserves user data across requests.
- Client-side Techniques:
- - ViewState: Stores values in hidden field on page.
- - HiddenField: Stores small amount of data in page.
- - QueryString: Appends data to URL.
- - Cookies: Store data in browser.
- Server-side Techniques:
- - Session: Stores user-specific data on server.
- - Application: Stores global data shared across users.
- - Cache: Stores frequently used data for fast retrieval.

Q12. What is Web.config file? Explain its advantages.

- Web.config is an XML-based configuration file in ASP.NET.
- Uses: Stores database connection strings, session settings, security settings, error handling policies.
- Advantages:
- - Centralized configuration management.
- - Changes do not require server restart.
- - Human-readable XML format.
- - Automatically detected by ASP.NET on modification.

Q13. Explain Web Services in ASP.NET.

- Web Services are components that allow communication between applications over HTTP using XML.
- Technologies: SOAP, XML, WSDL, UDDI.
- Advantages:
- - Language and platform independent.

- - Reusable across applications.
- - Based on open standards.
- Limitations:
- - Performance issues due to XML overhead.
- - Possible interoperability issues across platforms.

Q14. Explain Authentication and Authorization in ASP.NET.

- Authentication: Verifying identity of users.
- - Types: Forms Authentication, Windows Authentication.
- Authorization: Controlling access to resources after authentication.
- - Role-based and permission-based.
- Together they ensure secure access in web applications.

Q15. What is Caching in ASP.NET? Explain types.

- Caching improves performance by storing frequently used data in memory.
- Types of Caching:
- - Output Caching: Stores entire page output.
- - Data Caching: Stores specific data objects.
- - Application Caching: Stores data globally accessible.
- Advantages: Faster access, reduced database load, better scalability.